
Assignment 4 - Experimenting BERT on AG news

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1 Abstract

We investigate the performance of BERT on AG News, a topic classification dataset with four categories. Using both probing and fine-tuning approaches, we assess the effectiveness of BERT’s pretrained representations. In the probing setup, mean-pooled embeddings combined with logistic regression achieved 88.3 percent accuracy, outperforming [CLS] and last-token embeddings. Fine-tuning the full model improved accuracy to 92.7 percent, confirming the value of task-specific training. An untrained classification head, in contrast, defaulted to predicting a single class, achieving only 24.9 percent accuracy. Attention visualizations showed that correct predictions focused on semantically relevant tokens, while errors were associated with high attention weights on misleading or domain-neutral tokens. These results suggest that BERT captures strong semantic features even without task-specific tuning, but fine-tuning substantially improves performance and interpretability.

2 Introduction

Large Language Models (LLMs) like BERT (Devlin et al., 2018) have had a major impact on natural language processing by providing reusable text representations learned from large, unlabeled corpora. In this assignment, we explored how well BERT performs on AG News (Zhang et al., 2015), a four-class classification task involving short descriptions of news articles.

Our experiments were divided into two setups: probing and fine-tuning. In the probing setup, we froze BERT’s parameters and extracted sentence-level embeddings using three methods: the [CLS] token, the last token, and mean pooling. These embeddings were passed to lightweight classifiers (logistic regression and KNN) to evaluate the quality of BERT’s pretrained representations without further training.

For fine-tuning, we updated all of BERT’s weights on a subset of 10,000 training examples and evaluated on a 4,000-example validation set. This allowed BERT to learn from the dataset and adapt its internal representations to the classification task. We used the AdamW optimizer, dynamic padding, and a small batch size to accommodate limited GPU memory.

We also included a baseline using an untrained classification head, as well as visualizations of attention patterns from different BERT layers to gain insight into how the model distributes focus during correct and incorrect predictions.

Key findings from our experiments include:

1. **BERT’s frozen embeddings provide strong semantic representations:** Using mean-pooled embeddings and logistic regression yielded a test accuracy of **88.3%**, showing that BERT’s frozen representations already capture significant semantic information.
2. **Simple classifiers can be very effective when paired with good embeddings:** With mean-pooled embeddings and $K = 9$, KNN achieved a test accuracy of **88.0%**, confirming the effectiveness of simple models when paired with strong embeddings.
3. **Fine-tuning improves performance further:** Training the full model increased test accuracy to **92.7%** and macro-F1 to **0.927**, demonstrating the downstream benefits of letting BERT learn the specific task.
4. **The untrained classification head performs poorly:** The frozen BertForSequenceClassification model defaulted to predicting one class, achieving only **24.9%** accuracy and an F1 score of **0.10**, likely due to random initialization and lack of learned task-specific patterns.

5. **Attention visualizations reveal how BERT makes decisions:** Across multiple examples, we found that correct predictions tended to focus attention on semantically meaningful tokens, while errors often involved attention to irrelevant words. Comparing attention at layer 6 and layer 11 highlighted a shift from general or syntactic words to task-specific semantic cues, supporting prior work on BERT’s layered representations.

3 Datasets

We used the AG News dataset, a topic classification benchmark with news samples categorized into World, Sports, Business, and Science/Technology. Each sample includes a headline and a short description. The full dataset contains 120,000 training and 7,600 test samples. For fine-tuning, we used a reduced version with 10,000 training and 4,000 validation samples.

We loaded the dataset with the HuggingFace `datasets` library and tokenized it using `bert-base-uncased`, which lowercases text and applies WordPiece tokenization. Padding and truncation were handled automatically to fit within BERT’s 512-token input limit, so no additional preprocessing was needed.

Class distributions were nearly balanced across all splits, and token lengths ranged from 15 to 347 which is suitable for multi-class classification with BERT.

4 Benchmark and Methodology

We experimented with both probing and fine-tuning approaches using the `bert-base-uncased` model from HuggingFace’s Transformers library. This architecture consists of 12 transformer encoder layers, each with 12 self-attention heads and a hidden size of 768, totaling around 109 million parameters. The model was pretrained using masked language modeling and next sentence prediction on a large English corpus.

In the probing setup, all weights of the BERT model were frozen. We extracted fixed-length sentence embeddings from the final hidden layer using the following strategies:

- **CLS Token:** the output of the [CLS] token (position 0)
- **Last Token:** the embedding of the last non-padding token
- **Mean Pooling:** the mean of all non-padding token embeddings

These embeddings were then used to train lightweight classifiers (K-Nearest Neighbors (KNN) and multiclass logistic regression). KNN was evaluated for odd values of k from 1 to 29, with the best k selected based on validation accuracy. For logistic regression, we used scikit-learn’s implementation with the multinomial option and `max_iter=1000`.

To avoid recomputing embeddings for each run, we precomputed and cached them to disk using `pickle`. Embeddings were generated for the training, validation, and test sets, allowing us to compare classifier performance across different splits. We observed that mean-pooled embeddings generally led to the best results across both classifiers.

For the fine-tuning approach, we used `BertForSequenceClassification`, which adds a linear classification head on top of the [CLS] token output. We trained the model end-to-end on a subset of 10,000 training examples and validated on a separate set of 4,000 examples. The model was trained for three epochs with a batch size of 8 using the AdamW optimizer and a learning rate of 5×10^{-5} . Training was performed on McGill’s GPU server (Mimi), and dynamic padding was applied using a data collator to reduce unnecessary computation.

We evaluated model performance after each epoch using loss, accuracy, and macro F1 score on the validation set. However, due to logging constraints on the GPU server, we were only able to recover validation metrics from the first epoch. The final model was tested on the full 7,600-example test set. To better understand the model’s predictions, we also ran the fine-tuned classifier on several examples and analyzed attention scores from different layers.

5 Results

5.1 Probing Results

Table 1: Probing Results with Frozen BERT Embeddings

Embedding Type	Classifier	Best K	Test Accuracy	Val. Accuracy
First Token (CLS)	KNN	15	83.1%	82.3%
	Logistic Regression	-	88.1%	85.2%
Last Token	KNN	17	82.9%	83.0%
	Logistic Regression	-	87.3%	86.4%
Mean Pooled	KNN	9	88.0%	87.3%
	Logistic Regression	-	88.3%	86.7%

The results from probing with three different strategies (CLS token, last token, and mean pooling over non-padding tokens) are summarized in **Table 1**. Mean-pooled embeddings produced the best results across both classifiers. Logistic regression achieved a test accuracy of **88.3%**, while KNN reached **88.0%** with $k = 9$.

For the [CLS] token strategy, logistic regression achieved **88.1%** accuracy, and KNN reached **83.1%** with $k = 15$. Using the last non-padding token resulted in test accuracies of **87.3%** for logistic regression and **82.9%** for KNN with $k = 17$.

Overall, logistic regression consistently outperformed KNN across all embedding types.

5.2 Fine-tuning Results

The results from the fine-tuning process are as follows:

Model	Test Accuracy	F1 Score
Untrained (Frozen BERTForSequenceClassification)	0.249	0.1011
Trained Model	0.92724	0.92713

Table 2: Performance comparison of untrained vs trained BERT model

It is also interesting to note the scores after the first epoch of training: Training Loss: 0.34262, Validation Loss: 0.26806, and Validation Accuracy : 0.91350.

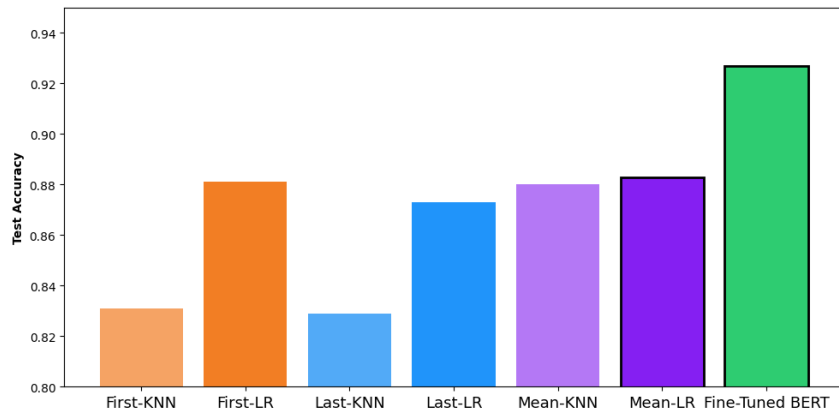


Figure 1: **Test accuracy comparison** of BERT probing strategies versus fine-tuning. The best probing approach (Mean-pooled embeddings with Logistic Regression, **88.3%**) is outperformed by fine-tuned BERT (**92.7%**).

Figure 1 shows the test accuracies for all probing strategies alongside the fine-tuned BERT model. The highest accuracy among the probing methods was 88.3%, achieved using mean-pooled embeddings with logistic regression. Fine-tuned BERT outperformed all probing variants with a test accuracy of 92.7%.

Below are some examples of our trained model’s performance, obtained from running it on the test dataset. We used these examples to generate our heatmaps, depicted later on in this section.

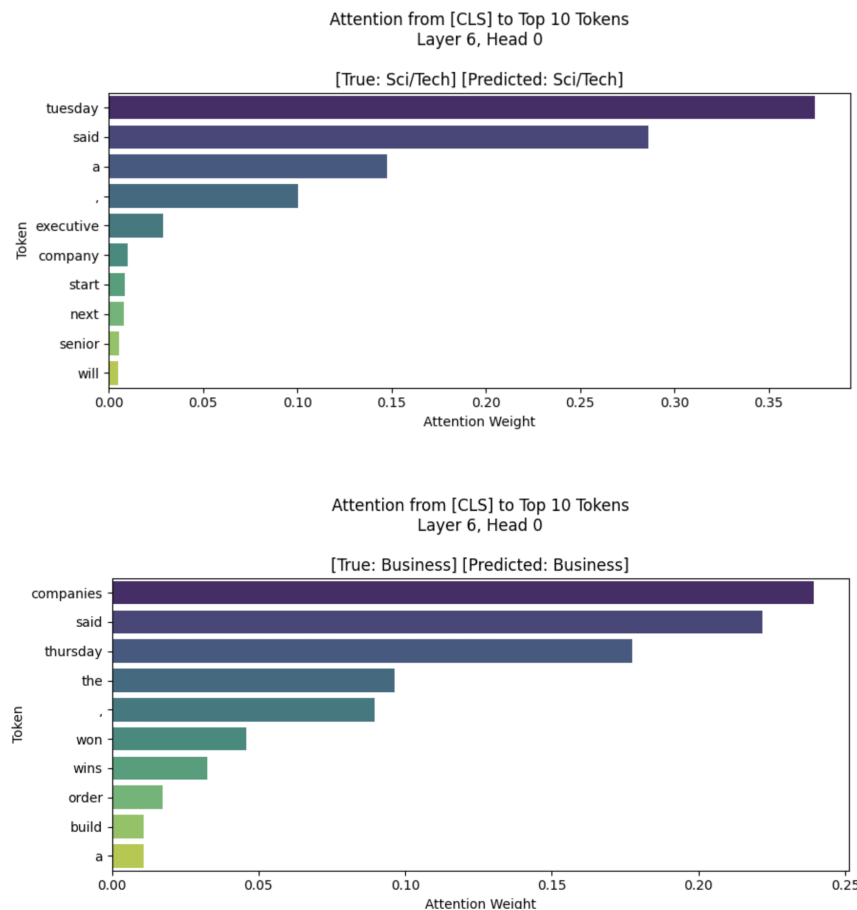
Correctly Classified Examples

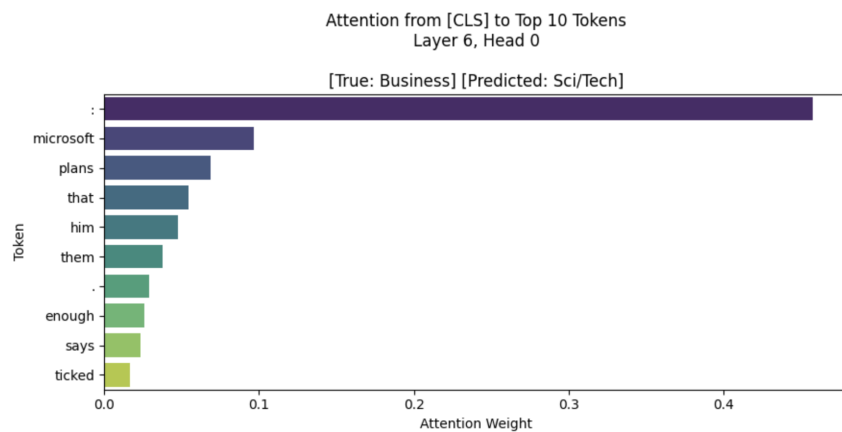
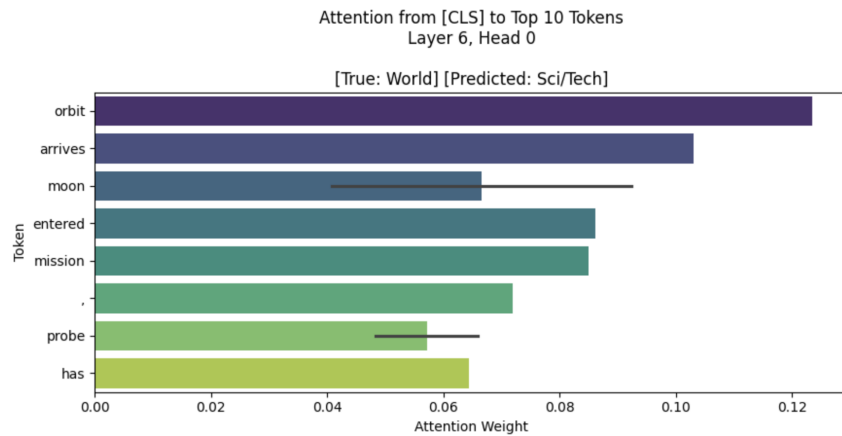
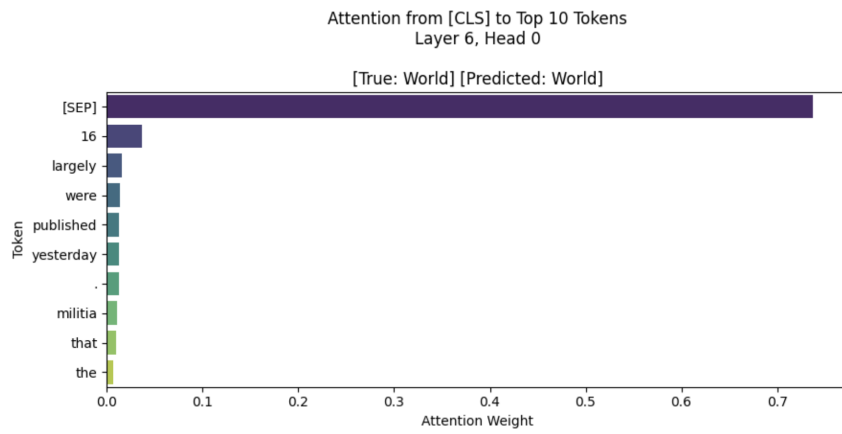
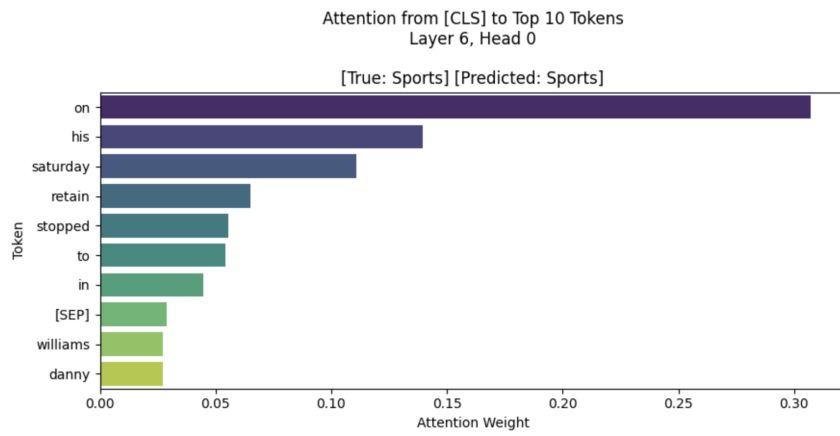
True Label	Predicted Label	Confidence	Text
Sci/Tech	Sci/Tech	0.99	Intel Rosedale chip brings WiMax closer. Intel Corp. has...
Business	Business	0.83	Alcatel wins \$1.7 billion network deal. PARIS - Alcatel, the...
Sports	Sports	0.99	Ukrainian Klitschko retains WBC heavyweight crown. Vitali Klitschko of Ukraine...
World	World	1.00	Sudan accused of sharing bases with militia. A human rights group...

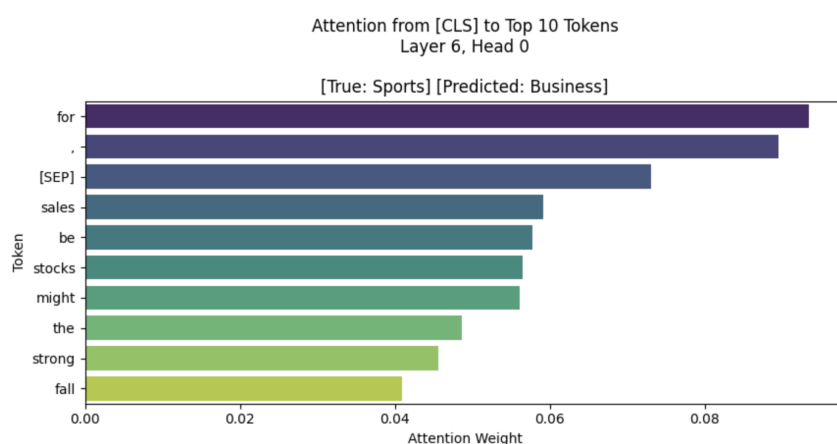
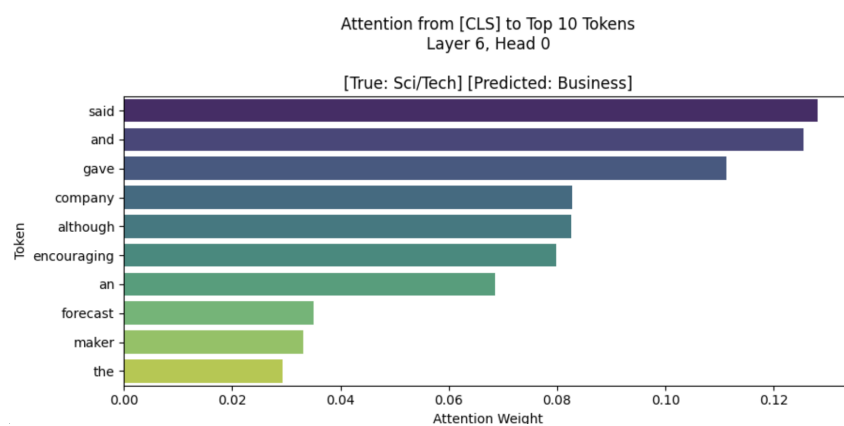
Incorrectly Classified Examples

True Label	Predicted Label	Confidence	Text
World	Sci/Tech	0.87	Europe probe arrives at the Moon. The Smart 1 lunar probe...
Business	Sci/Tech	0.81	Gates: Microsoft to Tackle the Spyware Problem...
Sci/Tech	Business	1.00	Truck maker delays report. Truck maker Navistar International...
Sports	Business	0.94	Auto Stocks Fall Despite Ford Outlook. Despite a strong profit...

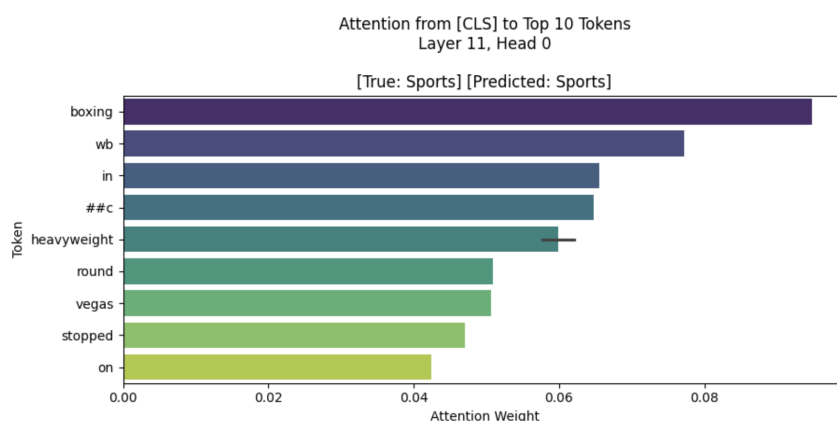
Below are the generated heatmaps for the attention from the CLS to the top 10 tokens for each topic, positive and negative. Due to the fact that the BERT-for-sequence-classification model we used had 11 layers, we chose to depict layer 6, head 0 for every example. This allowed us to view the attentions at a middle layer for each example. According to Clark et al., 2019 Heads from the middle layers may attend from [CLS] to important content words that inform the classification decision.

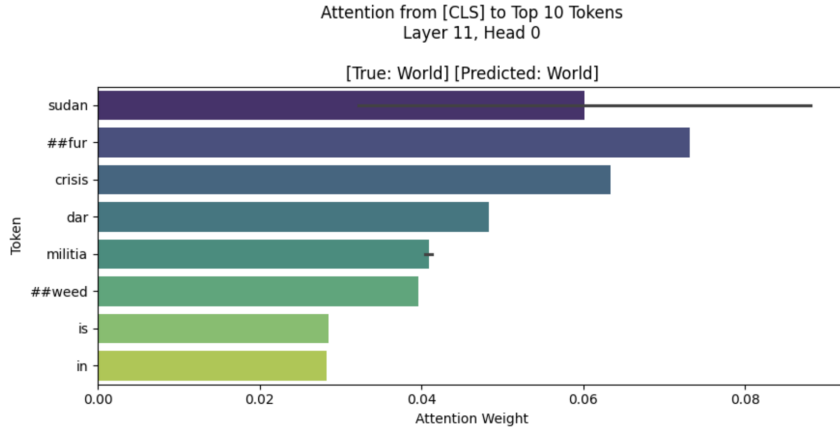
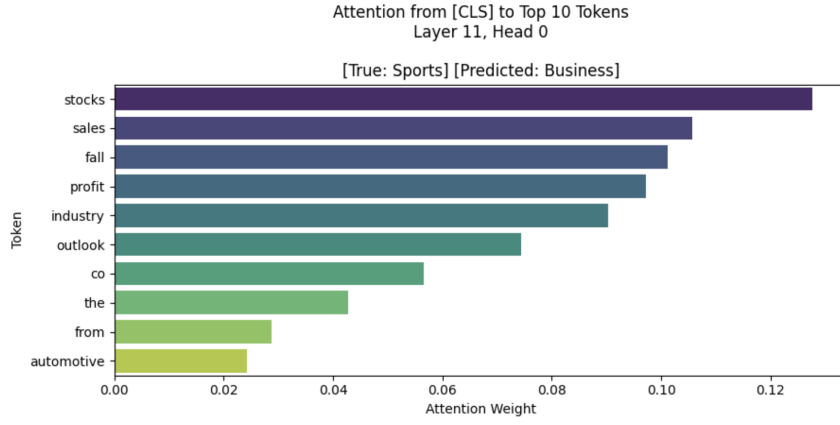






As discussed in prior research referenced in our introduction, different types of linguistic information are captured at different layers of a BERT model, with representations becoming increasingly abstract and semantic in the higher layers. To explore how attention varies across layers—particularly how the final layers emphasize different tasks than middle layers—we analyzed the attention weights from layer 11 (the final layer) across all examples. Below are some results from this experiment:





6 Discussion and Conclusion

6.1 Probing and Representation Quality

The results from probing with three different strategies (CLS token, last token, and mean pooling over non-padding tokens) showed that mean pooling consistently led to the best performance. The best configuration—mean pooled embeddings with logistic regression—reached a test accuracy of 88.3%. This supports prior work suggesting that BERT’s semantic information is spread across tokens and benefits from aggregation (Tenney et al., 2019; Clark et al., 2019). Across all probing setups, logistic regression outperformed KNN, indicating that the frozen embedding space is close to linearly separable for this classification task.

6.2 Fine Tuning and Model Performance

In our study, we fine-tuned the BERT model for sequence classification, specifically using the BertForSequenceClassification architecture, which applies a linear classification head on top of the pooled output of the base BERT encoder. Fine-tuning this model proved essential for achieving strong performance on our news classification task.

The contrast between the untrained and fine-tuned models highlights the importance of task-specific training. The frozen BertForSequenceClassification model performed extremely poorly, achieving only 24.9% test accuracy and an F1 score of 0.1011. Interestingly, it predicted almost all test samples as belonging to the “science/technology” class. This behavior can be explained by the fact that without training, the linear classification head remains randomly initialized, and its outputs are effectively dominated by the initial weight biases. Without learning, the model lacks the ability to discriminate between classes based on input content.

Once we fine-tuned the model, performance improved dramatically. After a single epoch of training on 10,000 examples, the model reached 91% validation accuracy. After three epochs, it achieved almost 93% validation accuracy and an F1 score of 0.927. These results illustrate that BERT’s pretrained layers, when paired with a fine-tuned classification head, can adapt well to classification tasks.

6.3 Analyzing Attention Across Layers

For each example, we visualized attention from head 0 of layer 6 and head 0 of layer 11, allowing for a direct comparison of intermediate versus deeper-layer attention.

We observed a clear pattern in the nature of the attended tokens. In layer 6, the most-attended words tended to be more general or syntactically functional (e.g., "on", "his", "Saturday"). In contrast, layer 11 attention focused on semantically rich, class-indicative words. For example, in a correctly predicted "sports" article, layer 11 attention highlighted terms like "boxing" and "heavyweight", which are highly relevant to the predicted label. Similarly, in an incorrectly predicted case (true label: sports, predicted label: business), the top tokens in layer 11 included "stocks", "sales", and "profit"—words likely responsible for the misclassification.

Our findings are consistent with prior work on the interpretability of BERT. Tenney et al. (2019) and Peters et al. (2018b) demonstrated that BERT processes linguistic information in a layered hierarchy, progressing from shallow syntactic features to deeper semantic understanding. According to Tenney et al., part-of-speech tags and constituents appear in lower layers, while dependencies, semantic roles, and coreference emerge in higher layers. This aligns with our observation that layer 6 attention tends to capture more syntactic or positional cues, while layer 11 captures richer, task-specific semantic information.

Overall, our experiments support the view that BERT builds up linguistic understanding layer by layer. Early and middle layers focus on syntax and surface-level structure, while upper layers integrate semantic content essential for tasks like news classification. It is, however, important to note that, while attention visualizations can offer valuable insights into BERT’s decision-making process, we should always interpret them cautiously: attention isn’t always explainable. In many cases, the tokens receiving the highest attention weights are not the ones we would intuitively expect the model to focus on. The model may assign certain weights/rely on features that are not easily interpretable, and may not be what we expect. As such, while attention maps can be valuable tools, their interpretations should be taken with a grain of salt.

6.4 Conclusion and Future Directions

The 4.4% performance gap between probing and fine-tuning highlights a tradeoff between efficiency and performance. While frozen embeddings paired with simple classifiers offer a strong and lightweight baseline, full fine-tuning unlocks additional accuracy gains at the cost of greater computation. Additionally, training on a 10,000-example subset may underestimate the full potential of BERT’s capacity. In future work, approaches like early stopping or partial unfreezing of BERT’s layers could help bridge the gap between probing and full fine-tuning while keeping training time reasonable.

7 Group Member Contributions

Negar implemented the code for the probing experiments. Victoria handled the fine-tuning code, attention visualizations, and wrote the results and discussion sections related to fine-tuning. Melek Deniz wrote the rest of the report, including the introduction, methodology, analysis of the probing results and prepared the tables and final formatting of the document.

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